



THE UNIVERSITY OF ARIZONA
ELLER COLLEGE OF MANAGEMENT

HSLopez School of
Business Analytics

CREDIT RISK ASSESSMENT AND FUTURE CREDIT EXTENSION DECISIONS

DECEMBER 11, 2024

BELLE HE, YIZHI CHEN, SRI TEJA, KONERU HARSHITHA
HSLopez School of Business Analytics

Table of Contents

<i>Introduction</i>	<i>1</i>
<i>Literature review</i>	<i>1</i>
<i>Data Overview and Preprocessing.....</i>	<i>2</i>
Section 1: Dataset Description.....	2
Section 2: Data Cleaning and Preprocessing	3
<i>Methodology and Analysis.....</i>	<i>4</i>
Section 1: Methodology	4
Section 2: Results and Analysis	5
Section 3: Key Insights.....	6
<i>Key Findings from Exploratory Data Analysis.....</i>	<i>7</i>
<i>Key Insights and Recommendations</i>	<i>7</i>
<i>Conclusion and Next Steps</i>	<i>11</i>
<i>Technical Appendix.....</i>	<i>0</i>
<i>Sources Consulted.....</i>	<i>4</i>

Introduction

Accurate credit risk assessment is a cornerstone of financial institutions' operations, directly influencing their stability, profitability, and compliance with global regulations. For instance, the global non-performing loan (NPL) ratio averaged 3.7% in 2022, resulting in billions of dollars in losses for banks worldwide (World Bank, 2022) ^[2]. These figures underscore the importance of robust evaluation mechanisms to assess borrowers' likelihood of default accurately. Beyond mitigating financial losses, effective credit risk management supports compliance with frameworks such as Basel III, which mandates banks to maintain sufficient capital reserves relative to their risk-weighted assets (Bank for International Settlements, 2021) ^[1]. Furthermore, accurate credit assessments bolster systemic financial stability by reducing the risk of cascading defaults, ensuring confidence in the global financial system.

The purpose of this report is to leverage data-driven analysis to improve credit management strategies. The focus is on enhancing risk mitigation, optimizing client segmentation, and informing future credit extension decisions. By analyzing key factors that differentiate "Good" clients from "Bad" clients, based on payment history and demographic attributes, this report aims to assist financial institutions in allocating resources effectively, reducing non-performing loans, and fostering stronger client relationships. This analytical approach not only enhances financial outcomes but also improves customer satisfaction and loyalty.

This report is structured to provide a comprehensive analysis and actionable recommendations. The first section introduces the dataset, outlining the preprocessing steps that ensured data quality and readiness for analysis. Following this, the methodology section elaborates on the techniques used, including exploratory data analysis (EDA) and statistical modeling approaches such as logistic regression and decision trees. The findings are presented with supporting visualizations, highlighting critical insights derived from the analysis. Lastly, actionable recommendations are provided to refine credit assessment processes, enhance risk management strategies, and optimize client targeting, accompanied by a technical appendix and references.

Literature review

Credit risk assessment is a pivotal function within financial institutions, aimed at evaluating the likelihood of a borrower's default. Over time, various methodologies have been developed to enhance the precision of these assessments. This review examines traditional statistical models, machine learning techniques, and hybrid approaches in credit risk evaluation.

1. Traditional Statistical Models

Historically, credit risk assessment has relied on statistical models such as logistic regression and discriminant analysis. These models utilize financial ratios and borrower characteristics to predict default probabilities. For instance, Altman's Z-score, developed in 1968, employs financial ratios to predict corporate bankruptcy with notable accuracy (Altman, 1968) ^[3]. Similarly, logistic regression has been widely adopted due to its simplicity and interpretability in estimating default likelihoods (Ohlson, 1980) ^[6].

2. Machine Learning Techniques

The advent of machine learning has introduced sophisticated algorithms capable of capturing complex, non-linear relationships in data. Techniques such as decision trees, random forests, support vector machines, and neural networks have been applied to credit scoring. For example, a study by Lessmann et al. (2015) compared various machine learning algorithms and found that ensemble methods, particularly random forests and gradient boosting machines, outperform traditional statistical models in predictive accuracy^[5].

3. Hybrid Approaches

Recent advancements have led to hybrid models that combine statistical methods with machine learning techniques to leverage the strengths of both approaches. Moradi and Mokhatab Rafiei (2019) proposed a dynamic credit risk assessment model integrating data mining techniques, demonstrating improved predictive performance over traditional models. Additionally, ensemble models that aggregate multiple classifiers have shown enhanced accuracy and robustness in credit risk prediction (Zhang et al., 2021)^[8].

4. Alternative Data and Network Analysis

The incorporation of alternative data sources, such as social network information, has emerged as a novel approach in credit risk assessment. Muñoz-Cancino et al. (2022) investigated the impact of social interaction features on creditworthiness assessment, revealing that network data can significantly enhance predictive performance, especially when traditional credit history is limited^[7].

5. Explainability and Regulatory Compliance

As models grow in complexity, ensuring their interpretability becomes crucial for regulatory compliance and stakeholder trust. El Qadi et al. (2021) addressed this by aligning machine learning model explanations with expert risk analysts' assessments, facilitating transparency in credit scoring models^[4].

In conclusion, the evolution of credit risk assessment methodologies reflects a shift from traditional statistical models to more complex machine learning and hybrid approaches. The integration of alternative data sources and the emphasis on model interpretability are shaping the future of credit risk evaluation, aiming to enhance predictive accuracy while maintaining transparency and compliance.

Data Overview and Preprocessing

Section 1: Dataset Description

The dataset includes information from two main sheets: application_record and credit_record, comprising a total of 36,457 unique client records. The data captures demographic, financial, and credit activity details necessary for evaluating credit risk.

Source and Size

Source: The dataset originates from a client database, containing both application details and monthly credit records.

Size: A total of 36,457 unique client IDs, with multiple attributes spanning demographic and financial characteristics.

Key Variables

1. Demographic Data (application_record):

CODE_GENDER: Gender of the client.

FLAG_OWN_CAR & FLAG_OWN_REALTY: Ownership status of a car and property.

NAME_FAMILY_STATUS: Marital status of the client.

CNT_CHILDREN & CNT_FAM_MEMBERS: Number of children and family size.

2. Financial and Employment Data (application_record):

AMT_INCOME_TOTAL: Total annual income of the client.

NAME_INCOME_TYPE: Income type (e.g., working, pensioner).

DAYS_EMPLOYED: Employment duration in days (negative values indicate employment; positive values indicate unemployment).

3. Credit Activity Data (credit_record):

MONTHS_BALANCE: Credit data starting month tracked backward from the current month.

STATUS: Credit repayment status, ranging from “paid off” (C) to “bad debts” (5).

Section 2: Data Cleaning and Preprocessing

1. Variable Exclusion

Chi-Square Test Results: Using the Kaiser-Meyer-Olkin test, variables such as **EDUCATION_TYPE**, **MARITAL_STATUS**, and **HOUSING_TYPE** were excluded as they showed slightly significant association with the target variable (Client.Status).

Excluded Variables: **FLAG_MOBILE** and **CNT_CHILDREN** were also dropped to avoid redundancy and multicollinearity issues.

2. Encoding Categorical Variables

Dummy Coding: The **NAME_INCOME_TYPE** variable was transformed into dummy variables for each income category (e.g., “Associate,” “Pensioner”).

Renaming for Clarity: Variables were renamed for readability, such as **INCOME_TYPECommercial associate** to **Associate**.

3. Standardization and Numeric Conversion

All columns were converted to numeric format to ensure compatibility with statistical and machine learning algorithms.

Numerical variables, such as **AMT_INCOME_TOTAL**, were standardized where necessary for analysis consistency.

4. Correlation Analysis

A correlation matrix was generated to identify multicollinearity among numeric variables.

Variables with high correlation coefficients were examined (≥ 0.80), and one of the pair was removed to avoid multicollinearity (CNT_CHILDREN & FAMILY_SIZE). For instance, Working was dropped as a baseline reference variable.

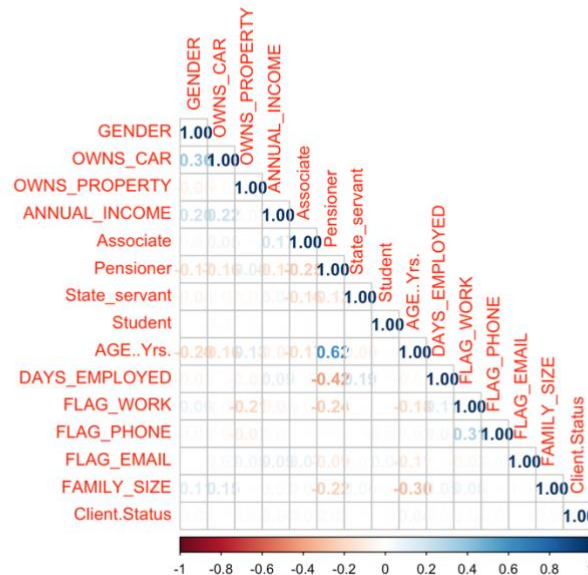


Fig. 1 Correlation Heatmap

5. Final Dataset

The final dataset included cleaned and transformed variables, with a structure optimized for statistical modeling and visualization.

Methodology and Analysis

Section 1: Methodology

This study employed a combination of regression models and machine learning algorithms to predict client status based on various demographic and behavioral attributes. A linear regression model was initially developed to understand the relationship between independent variables and the target variable, Client.Status. The coefficients of the linear regression provided insight into the significance and direction of each predictor. Following this, logistic regression and naïve Bayes classification algorithms were implemented to address the classification task, with special consideration given to handling class imbalance using SMOTE (Synthetic Minority Oversampling Technique).

A decision tree model was subsequently developed to enhance classification performance, particularly for imbalanced data. The decision tree, constructed using the rpart package with a complexity parameter of 0.0006, aimed to classify client status based on hierarchical decision

rules. Model evaluation was conducted on the test dataset, with results visualized through a confusion matrix. The decision tree achieved a predictive accuracy of 84.94%, demonstrating a strong bias toward the majority class. This underscored the limitations of the model in handling class imbalance, despite its overall accuracy.

Evaluation of the models' performance was conducted using a combination of metrics, including predictive accuracy, residual analysis for the regression models, and confusion matrices for the classification models. Multicollinearity among predictors in the linear model was assessed using Variance Inflation Factor (VIF), ensuring the robustness of coefficient estimates. The inclusion of the decision tree expanded the methodological toolkit, providing additional insights into the hierarchical structure of client status predictors.

Section 2: Results and Analysis

The linear regression analysis revealed statistically significant predictors, including Gender, Age, and Annual Income, among others, with the model explaining a minimal portion of the variance (Adjusted $R^2 = 0.0028$). Multicollinearity was found to be low, with all VIF values below the commonly accepted threshold of 10. However, the model's low R^2 indicated limited explanatory power, prompting the exploration of classification-based approaches.

The logistic regression model was developed on a SMOTE-adjusted training dataset to address class imbalance (original imbalance: 23205 "1" labels to 4138 "0" labels). The logistic model achieved a predictive accuracy of 42.07% on the test dataset, as determined from the confusion matrix, suggesting moderate alignment with observed outcomes. Predictors such as Annual Income and Age showed high statistical significance, reinforcing their influence on Client.Status.

Table 1. Confusion Matrix of Glm Model

Confusion Matrix of glm model		
	0	1
0	895	483
1	4797	2939

The naïve Bayes classifier was trained and evaluated, yielding probabilistic predictions for each client. When translated into discrete classifications, the model demonstrated a nuanced ability to handle the dataset's inherent imbalance. However, the accuracy of this model remains contextually dependent on thresholds and assumptions regarding prior distributions.

Table 2. Confusion Matrix of Naïve Bayes

Confusion Matrix of Naïve Bayes		
	0	1
0	1367	11
1	7673	63

The decision tree model, in contrast, achieved higher overall accuracy (84.94%) but showed limited sensitivity to the minority class due to imbalanced data. The confusion matrix revealed a substantial misclassification rate for minority class instances, highlighting the trade-off between overall accuracy and class-specific performance. This outcome emphasizes the need for further refinement, such as the integration of cost-sensitive learning.

Table 3. Confusion Matrix of Decision Tree

Confusion Matrix of Decision Tree		
	0	1
0	19	1359
1	14	7722

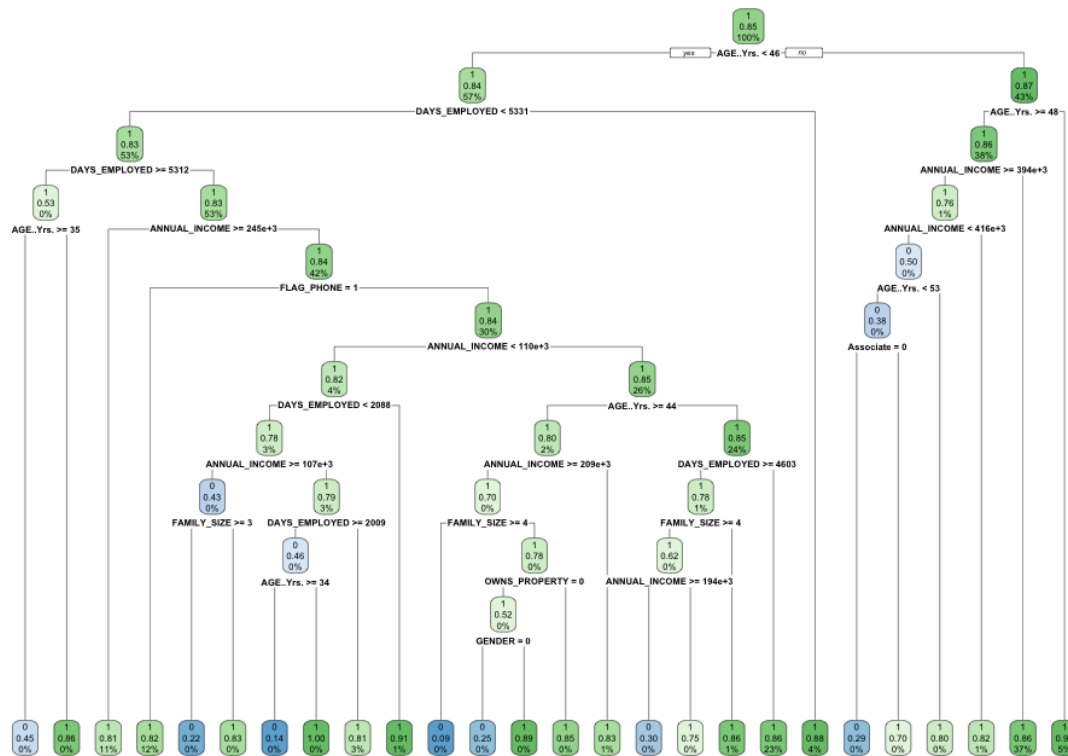


Fig. 2 Decision Tree Model Result

Comparing the approaches, logistic regression and naïve Bayes provided more actionable insights than linear regression due to their alignment with the classification nature of the problem. The decision tree model added interpretability through its rule-based structure, but highlighted challenges related to imbalance. Logistic regression, in particular, benefited from the application of SMOTE, enhancing its sensitivity to the minority class.

Section 3: Key Insights

The results underscore the importance of feature engineering and handling data imbalance in predictive modeling. While linear regression offered a foundational understanding, classification models such as logistic regression, naïve Bayes, and decision trees proved more suitable for predicting binary outcomes. Logistic regression emerged as a robust method, particularly when combined with oversampling techniques to address class distribution challenges. The naïve Bayes model's performance highlighted the utility of probabilistic predictions, especially in scenarios requiring nuanced decision thresholds. The decision tree model, while demonstrating

high overall accuracy, revealed the importance of mitigating class imbalance to achieve balanced predictive performance. These findings collectively inform strategies for improving client status prediction in future applications.

Key Findings from Exploratory Data Analysis

1. Income and Occupation Analysis

The “Working” category dominates the portfolio with 15,983 “Good Clients” and 2,836 “Bad Clients,” reflecting variability in repayment behavior. Conversely, “Pensioners” are largely low-risk, with a higher proportion of “Good Clients.” Other categories like “Commercial Associates” and “State Servants” show distinctive repayment behaviors. For instance, “Commercial Associates” exhibit higher default rates, while “State Servants” are a reliable segment. These variations underscore the importance of tailored risk assessments across occupation groups.

2. Educational Background

Educational attainment correlates strongly with repayment behavior. Secondary Education represents 67.96% of the client base, with a “Good-to-Bad” ratio of 85.03% to 14.97%. Higher education clients exhibit stable repayment trends with slightly better reliability, while Academic Degree holders stand out as a high-potential segment, boasting a 90.62% “Good Client” ratio and the highest average income.

3. Marital and Family Status

Married clients consistently perform better in repayment, whereas single individuals and those in civil marriages show elevated default rates. Clients with smaller family sizes and fewer dependents tend to exhibit better repayment behavior, highlighting financial strain as a significant factor among larger families.

4. Housing Type

Housing type revealed significant insights into repayment behavior. Clients living in “House / Apartment” were overwhelmingly categorized as “Good Clients,” demonstrating stable repayment patterns. Conversely, higher proportions of “Bad Clients” were observed among those living in “Municipal Apartments,” “With Parents,” or “Rented Apartments.” This underscores the impact of housing stability on financial reliability.

5. Repayment Behavior

Analysis of credit records revealed significant differences between “Good” and “Bad Clients.” Most “Good Clients” are categorized as “Paid Off” or “1–29 Days Overdue,” indicating strong repayment habits. Conversely, “Bad Clients” are concentrated in chronic delinquency categories (e.g., 90+ Days Overdue). Non-borrowing behavior (e.g., repeated “Status X”) was also noted as a predictive factor for defaults.

Key Insights and Recommendations

Model-Based Insights

The analysis highlights several critical factors influencing client credit behavior and areas for improvement in our predictive models. Key variables such as Annual Income, Age, and

Employment Duration significantly impact credit status. For example, clients with higher incomes are less likely to exhibit adverse credit outcomes, signaling an opportunity to refine risk evaluations by emphasizing these factors. Additionally, demographic traits like gender, car ownership, and property ownership influence credit behavior, suggesting potential to tailor marketing and product strategies to specific customer segments for improved engagement.

However, the analysis also identified limitations in model performance. While the decision tree model achieved an 84.93% accuracy rate, this figure is skewed by the model's tendency to predict the majority class (positive outcomes). Similarly, logistic regression showed a 42% accuracy, indicating a need to enhance the models with additional features or alternative techniques. Addressing these limitations requires rebalancing the dataset to reduce bias and ensure predictions are reliable for underrepresented groups, particularly those flagged as higher-risk.

Income Insights

The exploratory analysis revealed critical insights into income, educational attainment, and housing types, which directly affect credit behavior. Clients categorized under "Working" represented the largest income group, showcasing significant variability in repayment behaviors. Smaller but stable income groups, such as "Pensioners" and "State Servants," demonstrated consistent repayment reliability, offering opportunities for targeted financial products. The "Working" category forms most of the portfolio, with 15,983 Good Clients and 2,836 Bad Clients. Commercial Associates also show significant representation. Pensioners and State Servants demonstrate strong reliability, with balanced Good-to-Bad ratios, despite their smaller client volumes.

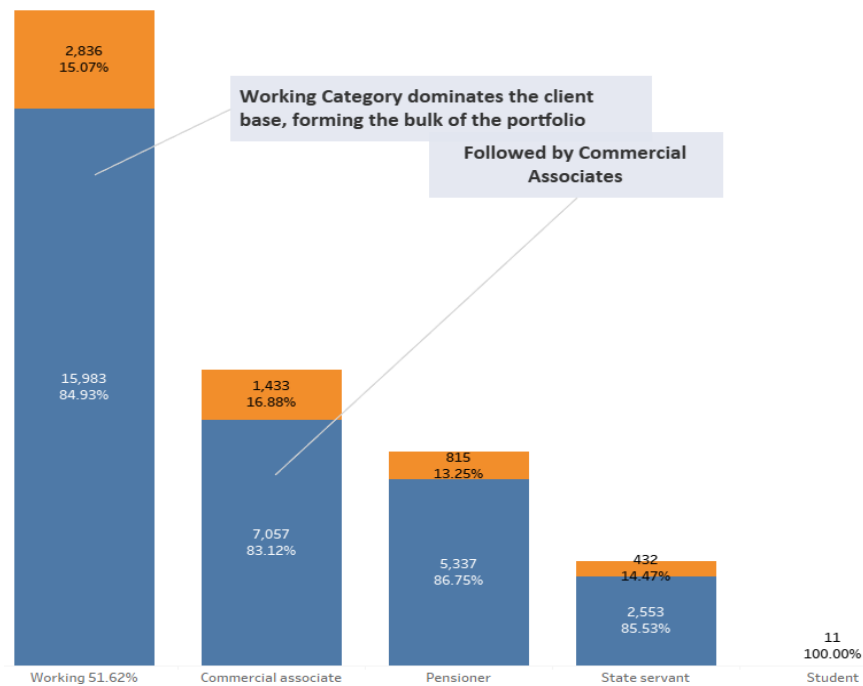


Fig. 3 Client Distribution by Income Type

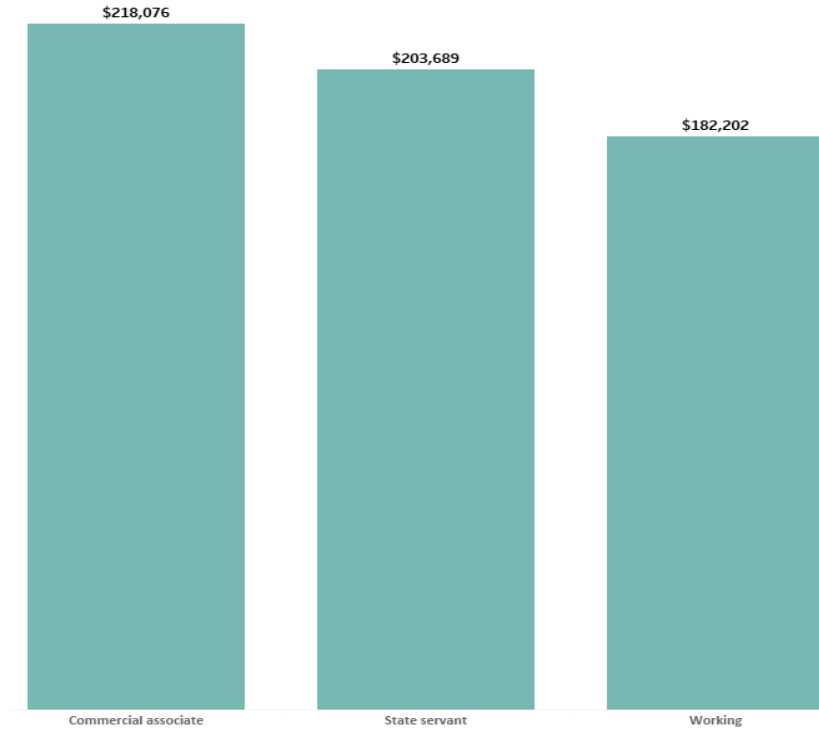


Fig. 4 Average Income by Income Type

Education Insights

Educational background significantly influenced creditworthiness. Clients with secondary education (67.96% of the total client base) formed the majority, showing reliable repayment behaviors. Meanwhile, Academic Degree holders (90.62% Good Clients) exhibited outstanding repayment reliability and the highest income levels, highlighting them as an ideal segment for premium credit offerings.

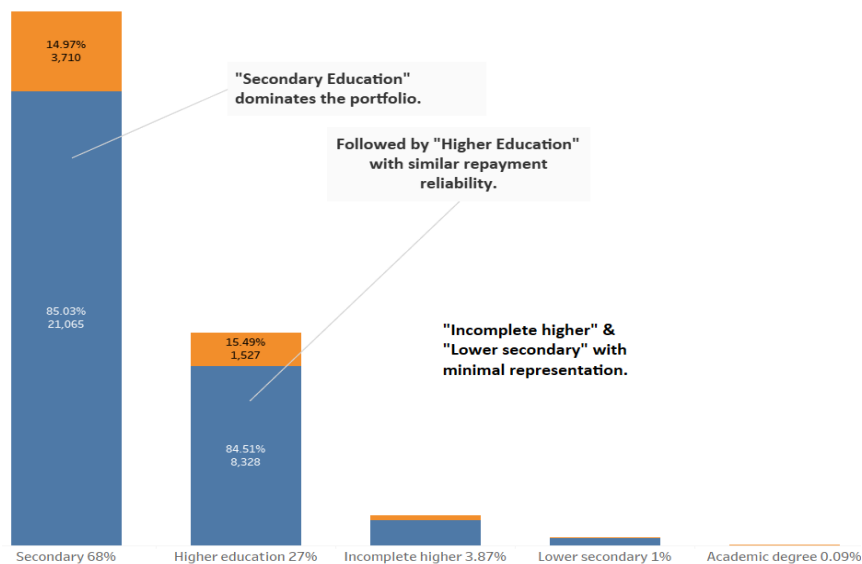
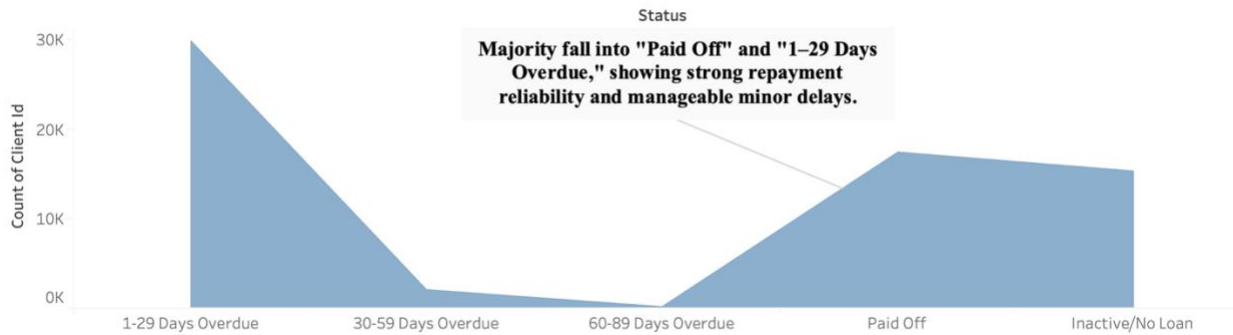


Fig. 5 Client Status by Education Type

Repayment Behavior Insights

The repayment patterns underscored notable distinctions between “Good” and “Bad Clients.” Most “Good Clients” were categorized as “Paid Off” or had minor delays (1–29 Days Overdue). In contrast, “Bad Clients” were heavily concentrated in high-risk categories such as “90+ Days Overdue,” reflecting chronic delinquency and disengagement.

Credit Activity Spread(good)



Credit Activity Spread(bad)

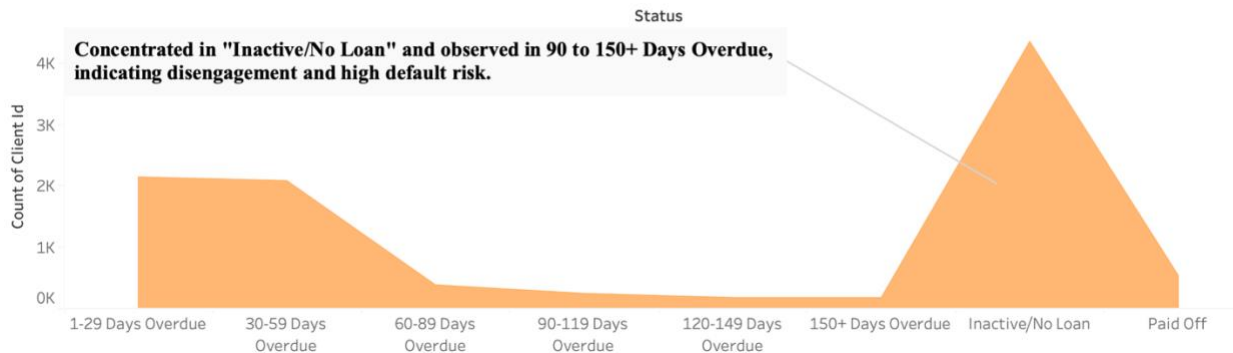


Fig. 6 Credit Activity Spread

Recommendations for Credit Management

1. Enhance Credit Scoring Models

Incorporate additional client characteristics, such as education level and housing type, to capture nuances in risk profiles more accurately. Adjust credit approval thresholds to reflect higher-risk factors, such as rented housing or lower educational attainment. This will ensure a more comprehensive risk evaluation process.

2. Mitigate Class Imbalances

Apply data rebalancing techniques like SMOTE to improve the fairness and accuracy of predictive models, particularly in representing minority client groups. This will reduce model bias and ensure better predictions for underrepresented segments.

3. Targeted Product Offers and Client Segmentation

Leverage demographic and employment insights to design tailored financial products for segments like pensioners, state servants, and students. These groups offer stable repayment potential and long-term relationship opportunities.

Develop premium credit options for high-income and high-reliability groups, such as clients with advanced educational backgrounds. Highlight these products through targeted marketing campaigns.

Expand credit offerings for high-potential groups like pensioners and state servants while introducing student-friendly credit plans to foster loyalty among younger clients.

4. Refine Marketing and Risk Management Strategies

Utilize predictive probabilities, such as those from Naive Bayes models, to prioritize outreach for clients with high creditworthiness. Implement early-warning systems to monitor clients at risk of delinquency, providing proactive interventions like payment reminders or support programs. For chronic defaulters (e.g., 90+ days overdue), offer debt restructuring or settlement options to minimize financial losses.

5. Develop Debt Recovery Initiatives

Establish specialized recovery teams to create customized repayment plans for high-risk clients. Improve communication channels to support repayment efforts, using legal action only as a last resort for persistent defaulters.

Conclusion and Next Steps

The analysis provided a comprehensive understanding of client demographics, income stability, and repayment behaviors, revealing critical insights for improving credit risk assessment. Key findings highlight the variability of repayment behavior among “Working” clients, the reliability of “Pensioners” and “State Servants,” and the influence of educational attainment and housing stability on creditworthiness. The results from predictive modeling reinforced these insights, with logistic regression and decision tree models offering actionable pathways for identifying at-risk clients and optimizing credit extension strategies. While the decision tree demonstrated the highest accuracy (84.94%), its sensitivity to minority classes remained limited, emphasizing the need for cost-sensitive enhancements.

Future analysis could explore alternative data sources, such as behavioral or social network data, to refine predictive accuracy further. Incorporating additional models like gradient boosting or hybrid techniques may address class imbalance more effectively. Enhanced segmentation strategies, targeting underrepresented groups such as Academic Degree holders, and personalized credit products for stable segments like “Pensioners” are key steps toward maximizing financial outcomes and client satisfaction.

To act on these recommendations, financial institutions should prioritize implementing dynamic credit scoring models, deploying early-warning systems for delinquency monitoring, and expanding offerings for reliable segments. For further details or collaboration, please contact Belle at beierhe@arizona.edu, Yizhi at chen25@arizona.edu.

Technical Appendix

Figure 1 – Confusion Matrix of glm regression model

Confusion Matrix of glm model		
	0	1
0	895	483
1	4797	2939

Figure 2 – Confusion Matrix of Naïve Bayes

Confusion Matrix of Naïve Bayes		
	0	1
0	1367	11
1	7673	63

Figure 3 – Decision Tree Model

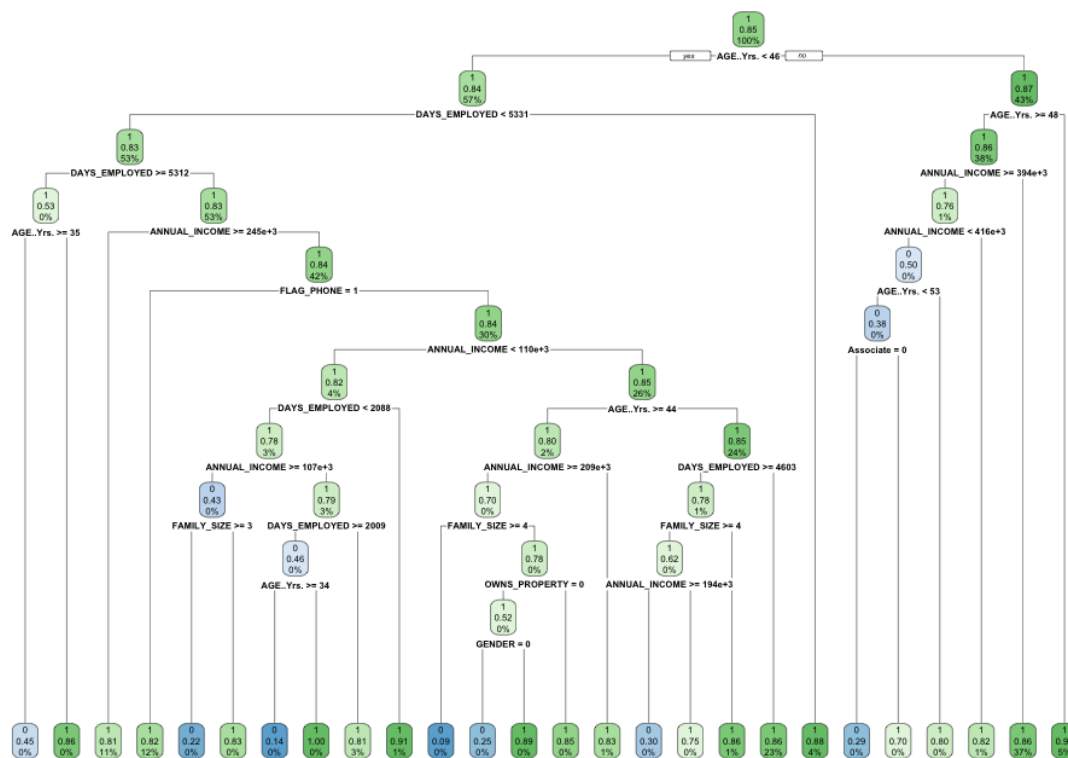


Figure 4 – Decision Tree Matrix

Confusion Matrix of Decision Tree		
	0	1
0	19	1359
1	14	7722

Figure 5 – Client Distribution by Income Type

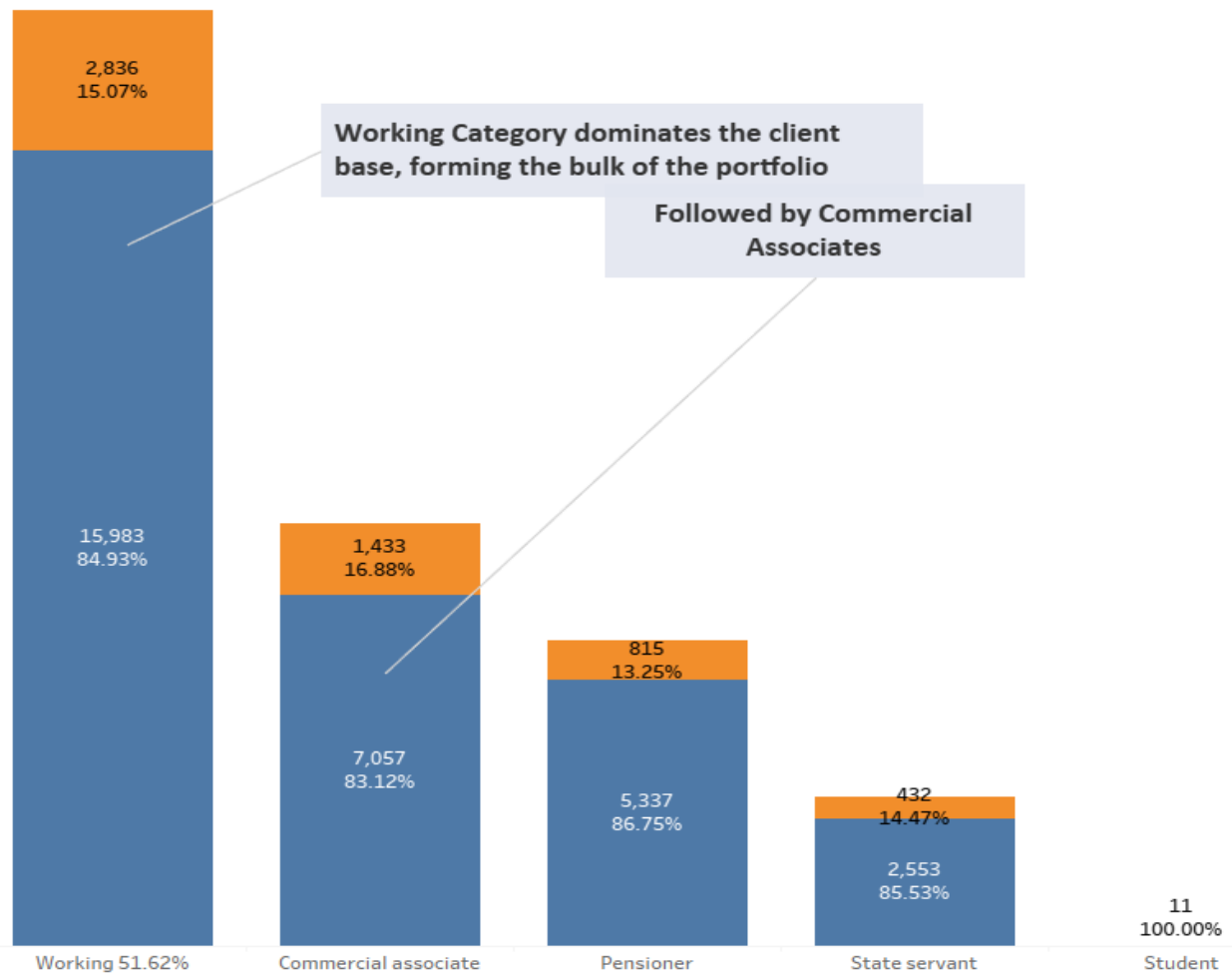


Figure 6 – Client Status by Education Type

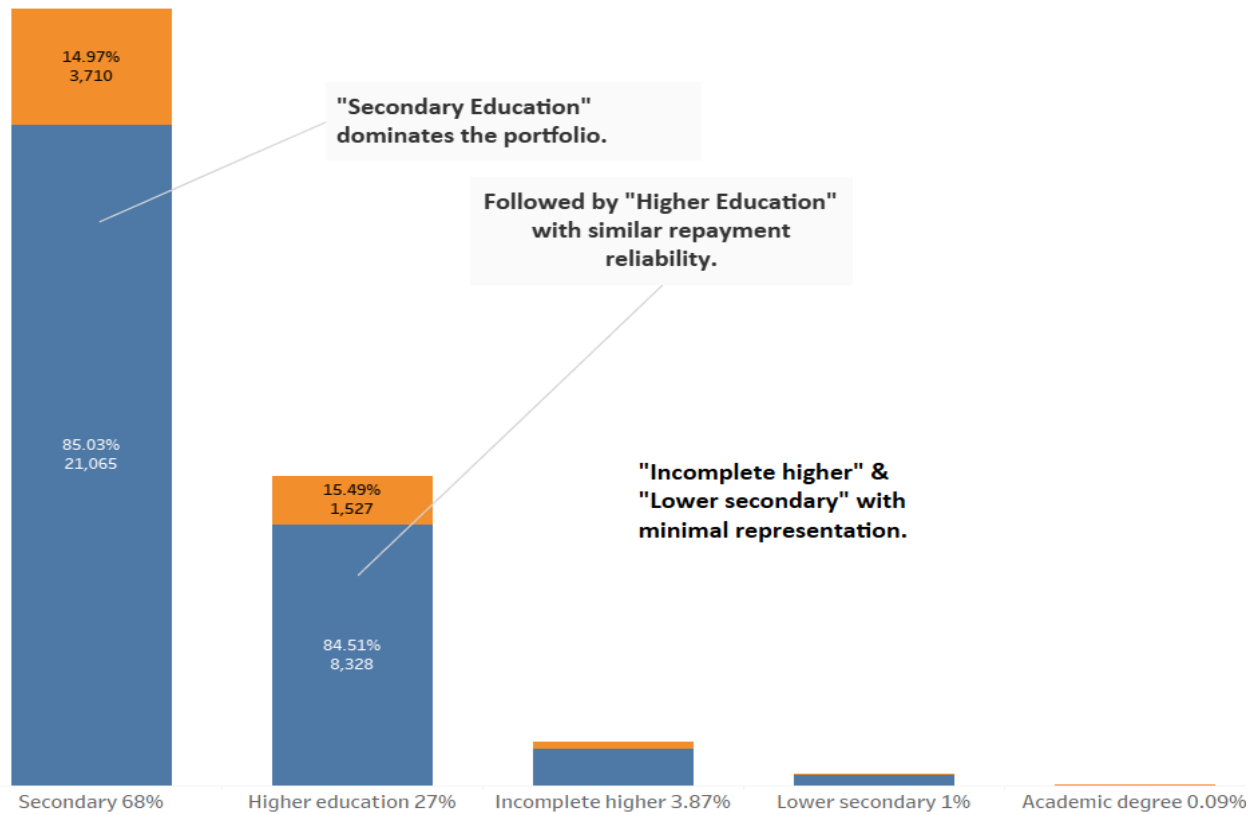
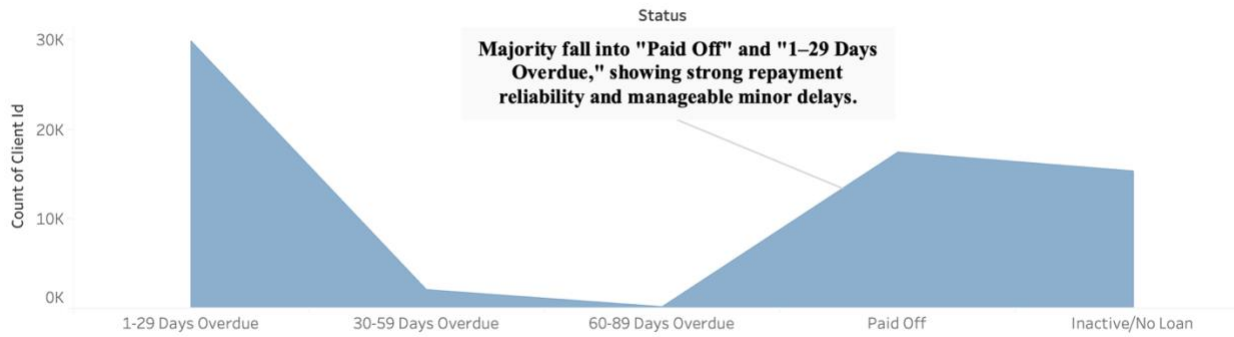
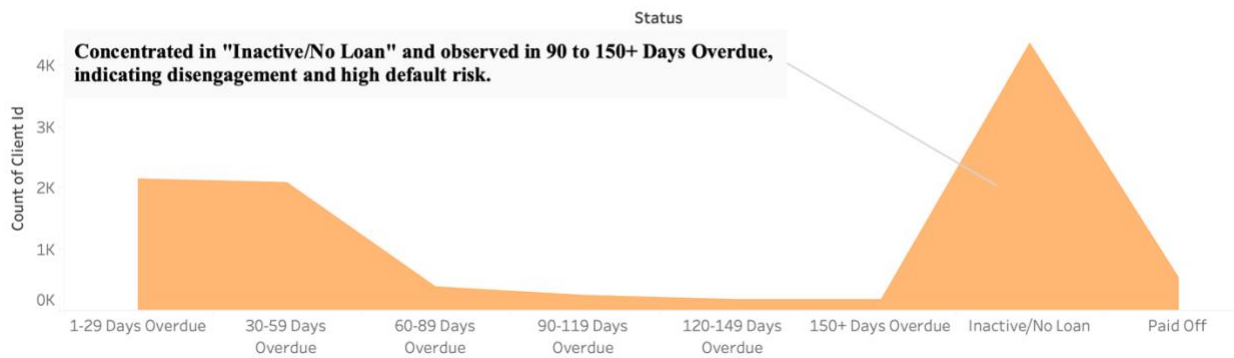


Figure 7 – Credit Activity Spread

Credit Activity Spread(good)



Credit Activity Spread(bad)



Sources Consulted

1. Bank for International Settlements. (2021). *Basel III: International regulatory framework for banks*. Retrieved from <https://www.bis.org>
2. World Bank. (2022). *Global non-performing loan ratios*. Retrieved from <https://www.worldbank.org>
3. Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4), 589–609.
4. El Qadi, A., Diaz-Rodriguez, N., Trocan, M., & Frossard, T. (2021). Explaining credit risk scoring through feature contribution alignment with expert risk analysts. *arXiv preprint arXiv:2103.08359*.
5. Lessmann, S., Baesens, B., Seow, H. V., & Thomas, L. C. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research*, 247(1), 124–136.
6. Moradi, S., & Mokhatab Rafiei, F. (2019). A dynamic credit risk assessment model with data mining techniques: Evidence from Iranian banks. *Financial Innovation*, 5(1), 1–27.
7. Muñoz-Cancino, R., Bravo, C., Ríos, S. A., & Graña, M. (2022). On the dynamics of credit history and social interaction features, and their impact on creditworthiness assessment performance. *arXiv preprint arXiv:2204.06122*.
8. Zhang, W., Yang, D., Zhang, S., Ablanedo-Rosas, J. H., Wu, X., & Lou, Y. (2021). A novel multi-stage ensemble model with enhanced outlier adaptation for credit scoring. *Expert Systems with Applications*, 165, 113872.